AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

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(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I096*

Team Name: *Dark Mod Devs*

Domain: *Digital Technologies, AI & Cyber Security*

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Problem Statement

Detecting and Disrupting Mule-Account Networks Behind Digital Payment Fraud

When someone is scammed on UPI, the money is split across a few ordinary people's accounts and withdrawn within minutes. Every bank sees only its own account, so nobody sees the whole chain in time to stop it.

Problem Understanding

- **Who:** everyday UPI users, especially first-time digital users such as parents, grandparents, small shop owners and daily wage workers. Students tempted to lend their account for commission are part of the problem too.
- **Cause:** UPI is instant and cannot be reversed, and each bank sees only its own account.
- **Effect:** complaints rose from 13.1 lakh (2023) to 24.02 lakh (2025), with Rs. 22,495 crore reported lost in 2025. *Source: MHA replies in Parliament, 2026.*

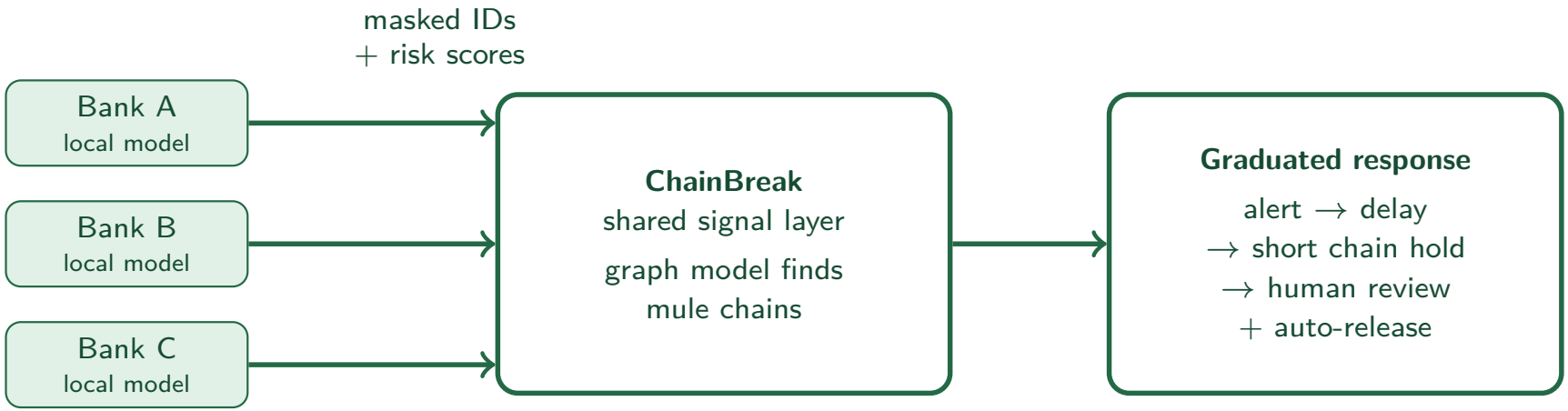
Existing Gaps & Challenges

- **Bank rule-based checks:** fixed rules, many false alerts, and each bank sees only its own accounts.
- **1930 helpline, CFCFRMS and I4C Suspect Registry:** action starts only after the victim reports, and the registry covers only identifiers already known to be bad.
- **RBI MuleHunter.AI:** strong at flagging single accounts, but as far as public information shows, it does not follow a live chain across banks.
- **Unsolved:** a real-time cross-bank view of the chain with a soft, reversible response.

Our Proposed Solution

- **Core idea:** *ChainBreak*, a shared signal layer where banks share only masked account IDs and risk scores. No customer data leaves a bank.
- **How it works:** a graph model finds mule chains (fast pass-through, many-to-one, one-to-many, brand new accounts). Federated learning lets banks improve it together without sharing raw data.
- **Response:** alert, small delay or extra confirmation, short hold on that chain only, then human review with an auto-release timer.
- **Value:** more of the victim's money is stopped midway, and honest users are delayed, not blocked.

Technical Approach



Banks train locally and send only masked signals; the shared layer links accounts across banks and triggers a gentle, reversible response.

Tech Stack

Frontend: React dashboard

Backend: Python, FastAPI

Database: Graph database, NetworkX

AI/ML: PyTorch Geometric, Flower (federated), hashing/masking

Innovation & Uniqueness

- **What is new:** it follows a money chain across banks in real time using only masked IDs and risk scores.
- **Why it suits the problem:** federated learning respects privacy and competition rules while every bank still contributes to one shared model.
- **Key differentiator:** detection by chain behaviour, so brand new mule accounts can be caught, plus a reversible response where every hold releases itself unless a reviewer confirms it.

Feasibility & Scalability

- **Feasibility:** the prototype needs only laptops, free open-source tools (Python, PyTorch Geometric, NetworkX, Flower, FastAPI, React) and synthetic transaction data, with a mentor or bank fraud officer to check our assumptions.
- **Scalability:** one local server at each bank plus one secure shared hub; later NPCI, a few banks and I4C as partners.
- **Risks:** real bank data cannot be shared, so results are on simulated data and need validation with partners. Wrong holds are limited by short, chain-only holds with auto-release and human review.

Cost for Product (Optional)

- Development cost: no software licence cost for the prototype (open-source tools, team laptops).
- Hosting cost: a cloud server (GPU optional) to train the graph model; amount depends on usage and will be scoped with partners.
- Estimated total: minimal for the prototype. Production cost to be estimated with NPCI and partner banks.

Use Cases

- **Use case 1, a first-time UPI user:** a scam payment is split across three mule accounts in different banks. ChainBreak links them, delays onward transfers and holds the chain, so most of the money is stopped before cash-out.
- **Use case 2, a student lending an account:** money passes through the account within seconds. The pass-through pattern raises the risk score, triggering extra confirmation, a short hold and a reviewer check.
- **Use case 3, a genuine shopkeeper on a festival day:** many small incoming payments look like a many-to-one pattern. The low score triggers only a small delay, and it releases automatically, so there is no freeze.

Impacts and Benefits

- **Users:** victims keep more of their money, honest users and shops are not blocked, and people trust UPI more.
- **Social and economic:** banks lose less and get fewer complaints, and police see the whole chain instead of pieces.
- **Measurable targets (on simulated data):** catch most mule chains before the last withdrawal, place a hold within about a minute of detection, and keep wrong holds on genuine accounts very low.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 16	Target 16.4	Peace, Justice and Strong Institutions	Disrupts illicit financial flows and gives police the whole chain
SDG 8	Target 8.10	Decent Work and Economic Growth	Builds trust in digital banking and protects honest users
SDG 9	Target 9.1	Industry, Innovation and Infrastructure	Makes payment infrastructure more resilient to fraud

Individual Member Contribution

Member	Contribution or responsibility
Thiruvenkadesuwaran S	Team leader: designs the slides and builds the prototype
K Prajit Raj	Research lead: collects fraud data and facts, studies RBI/NPCI rules and existing solutions
Nithin V	Solution and architecture lead: designs the cross-bank detection and holds, and makes the diagrams
Senthil Yohan N	Presenter: writes the pitch story, practises it and delivers it

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